

Carl J. Replogle

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Education

2026-	Doctor of Medicine (MD) University of Cincinnati College of Medicine, Cincinnati, OH
2022-2026	Bachelor of Science (BS) The Ohio State University, Columbus, OH Biomedical Engineering

Research Experience

2023-2026	The Ohio State University Undergraduate Research Assistant <i>McBride-Gagyi Lab - Bone Structure, Strength, and Repair</i>
2023-2024	Cincinnati Children's Hospital Medical Center Summer Undergraduate Research Fellowship <i>Whitlock Musculoskeletal Regenerative Medicine Lab</i>
2021-2022	Cincinnati SportsMedicine and Orthopaedic Center Clinical Research Assistant <i>Under mentorship of Dr. Samer Hasan</i>

Publications

1. Kurcz BP, DeBernardis D, Roberts A, Regala P, **Replogle CJ**, Fleckenstein CM, Nolan J, Hasan SS, Anatomic Total Shoulder Arthroplasty using a Short Humeral Stem and a Non-Augmented Minimally Cemented All-Polyethylene Glenoid: Minimum 2-Years Outcome and Predictors of Clinical Failure, *Seminars in Arthroplasty: JSES (2025)*, doi: <https://doi.org/10.1053/j.sart.2025.04.007>.

Presentations and Conference Abstracts

1. **Replogle, C.J.**, Diehl B., Barnes J., McBride-Gagyi, S. (March 2026). "In toto Mesh-Based Surface Segmentation of the Knee Joint for Quantitative MicroCT Analysis" [Poster], *Denman Undergraduate Research Forum*, Columbus, OH.
2. Barnes, J., **Replogle, C.J.**, Diop, K., Liu, Y., Klausner, J., Smith, A., Marshall, J., Walter, B., Schilaty, N., Bates, N., McBride-Gagyi, S., Purmessur, D. (December 2025). "Lower Limb Joint Degeneration Following Intervertebral Disc Injury and Force-Based Manipulation in a Mouse Model" [Poster], *Orthopedic Research Society 2026 Annual Meeting*, Charlotte, NC.
3. Robert Smallwood, Trinity Dort, **CJ Replogle**, Jacob Barnes, Kell Sprangel, Jake Willows, Magdalena Blaszkiewicz, Sarah Peters, Nicholas Greco, Kristy Townsend, Sara McBride-Gagyi, "Effects of Estradiol (17-alpha) on Osteoarthritis and Bone Development in Genetically Heterogeneous Het3 Male Mice." [Poster] *Orthopedic Research Society Annual Meeting*; February 10th, 2025

4. **Replogle CJ**, Smallwood R, Dort T, Sprangel K, Willows J, Peters S, Nicolas Greco, Townsend K, McBride-Gagyi S “Effects of Estradiol (17- α) on Osteoarthritis and Bone Development in Genetically Heterogenous Het3 Male Mice” [Poster] *Engineering in Healthcare: Industry and Research Symposium*; January 25th, 2025; Columbus, OH.
5. **Replogle CJ**, Ghorai SK, Gangavaram A, Whitlock PW “Fabrication of Patient Specific Bioinductive 3D Printed Scaffold for Osteochondral Defect Repair” [Poster] *Cincinnati Children’s Hospital Medical Center Summer Undergraduate Research Fellowship Capstone Poster Session*; July 31st, 2024; Cincinnati, OH.
6. S Hasan, C Fleckenstein, **C Replogle**, A Roberts, D Debernardis, B Kurcz “Anatomic Total Shoulder Arthroplasty Using a Short Humeral Stem and a Non-Augmented Minimally Cemented All-Polyethylene Glenoid: Minimum 2-Years Outcome and Predictors of Clinical Failure.” [Abstract] *15th International Congress on Shoulder and Elbow Surgery*; September 5-8th, 2023; Rome, Italy.
7. **Replogle CJ**, Ghorai SK, Henkel W, Whitlock PW “Biofabrication of Osteochondral Scaffolds Containing Embedded Decellularized Matrix-PLA Microspheres.” [Poster] *Cincinnati Children’s Hospital Medical Center Summer Undergraduate Research Fellowship Capstone Poster Session*; August 2nd, 2023; Cincinnati, OH.
8. **Replogle CJ**, Fleckenstein C, Hasan SS, “Early Outcome Following Anatomic Total Shoulder Replacement Using Short Stem Humeral and Minimally Cemented Glenoid Components.” [Talk and Poster] *Summit Country Day School Science Research Institute Colloquium*; February 28th, 2022; Cincinnati, OH.

Awards/Honors

2024-2025	The Ohio State University Honors Research Scholarship
2024	Kettering Biomedical Engineering Scholarship
2022-2026	Maximus Scholarship
2022	GE Star Award

Professional Organizations

2022-2026	Biomedical Engineering Society
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